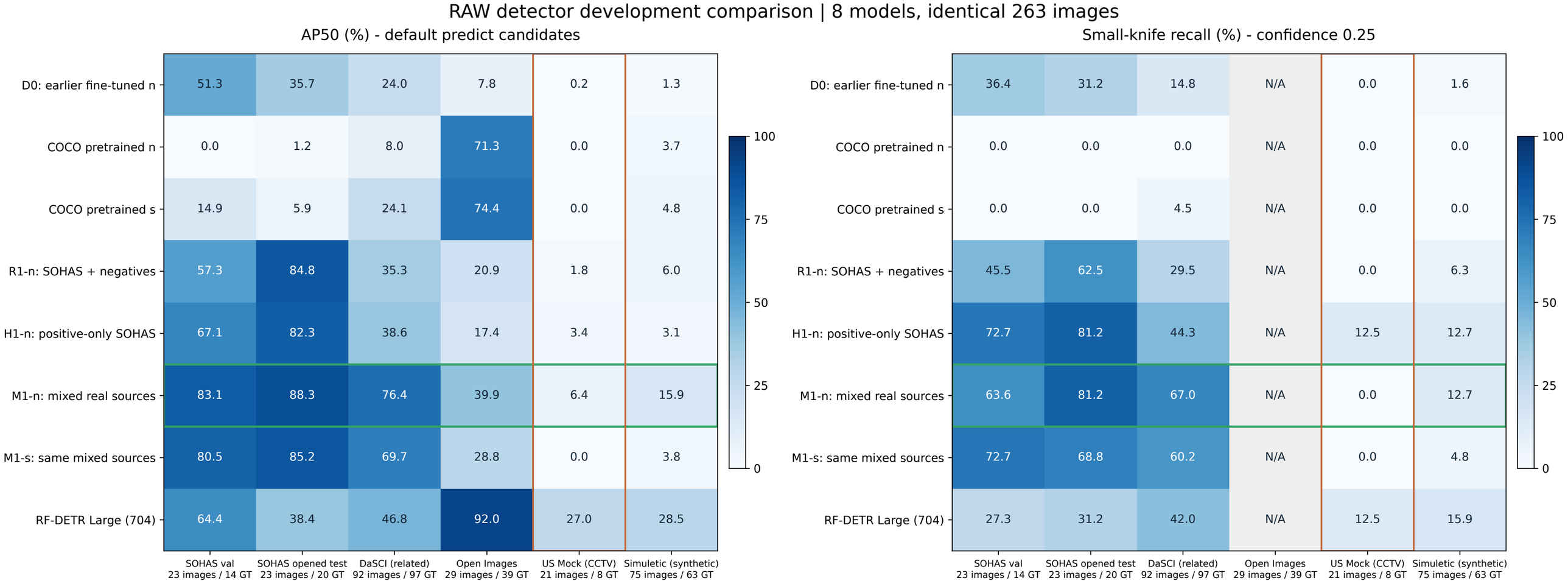
RAW detector development comparison | 8 models, identical 263 images
AP50 (%) - default predict candidates
D0: earlier fine-tuned n
COCO pretrained n
COCO pretrained s
R1-n: SOHAS + negatives
H1-n: positive-only SOHAS
M1-n: mixed real sources
M1-s: same mixed sources
RF-DETR Large (704)
51.3 35.7 24.0 7.8 0.2 1.3
0.0 1.2 8.0 71.3 0.0 3.7
14.9 5.9 24.1 74.4 0.0 4.8
57.3 84.8 35.3 20.9 1.8 6.0
67.1 82.3 38.6 17.4 3.4 3.1
83.1 88.3 76.4 39.9 6.4 15.9
80.5 85.2 69.7 28.8 0.0 3.8
64.4 38.4 46.8 92.0 27.0 28.5
100
75
50
25
0
SOHAS val
23 images / 14 GT
SOHAS opened test
23 images / 20 GT
DaSCI (related)
92 images / 97 GT
Open Images
29 images / 39 GT
US Mock (CCTV)
21 images / 8 GT
Simuletic (synthetic)
75 images / 63 GT
Small-knife recall (%) - confidence 0.25
D0: earlier fine-tuned n
COCO pretrained n
COCO pretrained s
R1-n: SOHAS + negatives
H1-n: positive-only SOHAS
M1-n: mixed real sources
M1-s: same mixed sources
RF-DETR Large (704)
36.4 31.2 14.8 N/A 0.0 1.6
0.0 0.0 0.0 N/A 0.0 0.0
0.0 0.0 4.5 N/A 0.0 0.0
45.5 62.5 29.5 N/A 0.0 6.3
72.7 81.2 44.3 N/A 12.5 12.7
63.6 81.2 67.0 N/A 0.0 12.7
72.7 68.8 60.2 N/A 0.0 4.8
27.3 31.2 42.0 N/A 12.5 15.9
100
75
50
25
0
SOHAS val
23 images / 14 GT
SOHAS opened test
23 images / 20 GT
DaSCI (related)
92 images / 97 GT
Open Images
29 images / 39 GT
US Mock (CCTV)
21 images / 8 GT
Simuletic (synthetic)
75 images / 63 GT
All images are opened development data; not independent final CCTV evaluation. Small = normalized bbox area <= 2%.
DaSCI is related source (one proxy group); synthetic data is separate. Open Images has no small GT. No event/GPIO/Orin speed claim.